



HSE COMPLIANCE EXTRACTION & REPORTING SYSTEM

Business Requirements Document (BRD) - Automated Extraction & PDF Reporting Module

DKSH HSE Compliance Division

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BUSINESS REQUIREMENTS DOCUMENT (BRD)

HSE CAPA Production System: Report Extraction Feature

Document Control

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1. Executive Summary & Project Context

1.1 Project Background

The DKSH HSE CAPA Production System is an enterprise-grade Health, Safety, and Environment (HSE) management platform designed to track, audit, and analyze workplace incidents, safety inspections, non-conformances, and Corrective and Preventive Actions (CAPA).

To maintain regulatory compliance (such as ISO 45001 and ISO 14001), support global corporate governance, and drive continuous safety improvements, HSE stakeholders require a robust, flexible, and secure Report Extraction Feature. This feature allows users to filter, compile, preview, and download critical safety performance records and CAPA logs.

1.2 Objective of the Feature

The primary objective of the Report Extraction Feature is to provide seamless, self-service data export capabilities. It translates raw operational HSE records such as working hours, incident tallies, CAPA ticket lifecycles, and audit results into auditable, executive-ready offline formats (Excel/CSV and PDF Executive Summaries).

2. User Roles & Security Matrix (Role-Based Access Control)

The Report Extraction feature must strictly enforce the existing role hierarchy to ensure data privacy, commercial safety security, and operational integrity.

Role Profile	Data Access Scope	Allowed Export Formats	Functional Permissions /
Superuser *(Global Admin)*	**Global Scope** Access to all countries, markets, business units, and historical archives.	ï¿½ Raw CSV/XLSX Dumps ï¿½ PDF Executive Summaries ï¿½ Board-level Slide Deck Exports	ï¿½ Full unrestricted download capacity. ï¿½ Access to cross-country labor statistics and incident investigations. ï¿½ Administrative control to modify master configurations (Distribution Centers) globally.
Regional HSE Manager *(Regional Coordinator)*	**Regional Scope** Access to all assigned countries/markets within their region (e.g., APAC).	ï¿½ Raw CSV/XLSX Dumps (Consolidated or by Market) ï¿½ PDF Executive Summaries	ï¿½ Full viewing, analytical, and extraction rights across all regional markets. ï¿½ Can verify investigations and CAPA sign-offs globally. ï¿½ Excluded from master master list (DC) modifications outside authorization.
Country HSE Manager *(Level 2)*	**Single Market Scope** Access restricted to their authorized Country/Market context.	ï¿½ Raw CSV/XLSX Dumps ï¿½ Market PDF Summaries	ï¿½ Restructured to their specific market (e.g. Thailand, Malaysia, Vietnam). ï¿½ Cannot export cross-border statistics without explicit global delegation.
Reporter *(Level 1)*	**Restricted Local Scope** Read-only access to localized dashboards and standard incident records.	ï¿½ Basic Excel lists of logged incidents and tickets.	ï¿½ Excluded from raw labor hour exports or aggregate payroll safety calculations. ï¿½ Data is masked for sensitive personal information (PII).

3. Scope of Report Extraction

3.1 In-Scope Requirements

- Multi-Format Export Engine: Support for CSV (comma-separated values), Excel-compatible spreadsheets, and PDF documents.
- Interactive Filter Panel: Granular parameters for Date Range (Quick Presets and Custom Calendar), Country/Market, Incident Classifications, Severity Level, and CAPA Status.
- DKSH Performance Indicators (KPIs): Calculation of safety indices, including:
 - Lost Time Injury Frequency Rate (LTIFR)
 - Total Recordable Injury Rate (TRIR)
 - Severity Rate (SR)
 - Work Hours vs. Incident Correlation Graphs

- 4. CAPA Roster Logs: Clear spreadsheets detailing Outstanding Actions, Owners, Root Cause Classifications, Target Closure Dates, and Aging Timelines.
- 5. Personal Identifiable Information (PII) Protection: Redaction of worker names, witness names, and medical details for standard reporter exports.

3.2 Out-of-Scope Requirements

- 1. Automated SFTP Delivery: Automatically pushing generated files to external third-party servers (to be considered for Phase 2).
- 2. Custom SQL Query Builder: Direct database querying capability for end-users. All exports must adhere to standardized system templates.

4. Functional & Interface Specifications

4.1 Filter and Criteria Definition

Before extracting any data, the user must interact with an intuitive, mobile-responsive filter interface.

🔗📅 Date Range Selector:

- Quick Presets: "Current Month", "Last 30 Days", "Current Quarter", "Year to Date (YTD)", and "Previous Fiscal Year".
- Custom Range: Dual interactive calendar widgets supporting Start Date and End Date.

🔗🌐 Market Scope Selection:

- If the user is a *Superuser*, this is an active dropdown displaying all active markets (e.g., Vietnam, Malaysia, Thailand, Singapore, Indonesia).
- If the user is a *Country HSE Manager* or *Reporter*, this dropdown is permanently disabled/locked and pre-selected to their authorized country.

🔗🚨 Incident Severity Filters: Checkboxes for:

- LTI (Lost Time Injury)
- MTC (Medical Treatment Case)
- FAC (First Aid Case)
- Near Miss
- Unsafe Act / Unsafe Condition

🔗🔄 CAPA Ticket Lifecycle Status: Options to filter by "All", "Open & Overdue", "Under Review", or "Closed & Verified".

4.2 Data Layout & Export Templates

Template A: Raw Operational Dump (CSV / XLSX)

Designed for deep data analysis, pivoting, and loading into corporate BI platforms (e.g., Power BI, Tableau).

Column Name	Data Type	Sample Value	Business Logic / Validation
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Ticket_ID	String (Unique)	`CAPA-2026-049`	Unique primary key formatted as standard ticket nomenclature.
Market_Country	Option	`Thailand`	Country where the incident or audit occurred.
Logged_Date	Date	`2026-04-12`	Timestamp of entry into the system (YYYY-MM-DD).
Incident_Classification	String	`LTI`	Classification hierarchy validated against safety standards.
Brief_Description	String	`Forklift collision near dock 4`	Summarized text entry (truncated to 255 characters for CSV safety).
Immediate_Action	String	`Quarantined forklift, area flagged`	Immediate mitigative action taken within 24 hours.
CAPA_Action_Required	String	`Reroute pedestrian walkway & repaint`	Permanent corrective action planned.
Action_Owner	String	`Somsak P.`	Designated assignee.
Target_Due_Date	Date	`2026-05-30`	SLA target date for completion.
Current_Status	Enum	`Overdue`	Active status: `Open`, `Closed`, `Overdue`.
Days_Open_Aging	Integer	`45`	Calculated as: `Current_Date` minus `Logged_Date` (if status is Open/Overdue).

Template B: Executive PDF Dashboard Summary

Designed for distribution during Monthly Country Management Meetings and Regional Safety Committee Reviews.

📄 Structure of PDF Document:

- Corporate Branding Header: DKSH Logo, Title "HSE Performance & CAPA Compliance Report", Selected Date Range, Generation Timestamp.
- Executive Summary Grid (KPI Highlights):
 - Safe Work Days: Consecutive days without a Lost Time Injury (LTI).
 - LTIFR (Lost Time Injury Frequency Rate): (Total LTIs * 1,000,000) / Total Working Hours.
 - TRIR (Total Recordable Injury Rate): (Total Recordable Cases * 1,000,000) / Total Working Hours.
- High-Level Visual Charts:
 - *Bar Chart*: Incidents by Type (LTI, MTC, FAC, Near Miss).
 - *Pie Chart*: CAPA Tickets status distribution.
 - *Trend Line*: Man-Hours vs Incident rates month-over-month.
- Overdue CAPA Action Table: Truncated roster of top 5 critical overdue items requiring immediate executive intervention.
- Formal Footer: "Page X of Y 📄 Confidential 📄 Generated via DKSH HSE CAPA Production Portal".

5. Calculations & Analytical Business Logic

To maintain mathematical standardization across all global operations, the export and analytical engine must calculate metrics on-the-fly according to the standard corporate rate formula.

5.1 Standard Formula Rate Principle

The system applies a uniform frequency rate standard based on 1,000,000 exposure hours worked:

$$\text{Standard Rate} = \frac{\text{Total Incident Cases of Specific Type}}{\text{Total Labor Hours Worked}} \times 1,000,000$$

5.2 Complete Mathematical Indicator Matrix

The safety rates and counts are mathematically classified and audited against the following master matrix:

No.	Indicator / Chart	Formula / Calculation	Short Form	Category Scope
1	**Fatalities Result**	$\frac{\text{Total Fatalities (No.7)}}{1,000,000} \times \text{Total Labor Hours Worked}$	**DEATH**	Fatal occurrences during work hours.
2	**High Consequence Work-Related Injuries**	$\frac{\text{Total High-Consequence Work-Related Injuries (No.8)}}{1,000,000} \times \text{Total Labor Hours Worked}$	**>6M**	Injuries resulting in recovery time >6 months.
3	**Lost Time Injury Rate**	$\frac{\text{Total Lost Time Injuries (No.9)}}{1,000,000} \times \text{Total Labor Hours Worked}$	**LTI**	Injuries causing at least 1 full day/shift of absence.
4	**Recordable Injuries**	$\frac{\text{Total Recordable Work-Related Injuries (No.8-13)}}{1,000,000} \times \text{Total Labor Hours Worked}$	**REC**	Sum of recordable injuries (No. 8 through 13).
5	**Work-Related Injuries**	$\frac{\text{Total Work-Related Injuries (No.8-14)}}{1,000,000} \times \text{Total Labor Hours Worked}$	**WRI**	Sum of all work-related injuries (No. 8 through 14).
6	**Property Damage Cases**	$\text{Absolute Case Count}$	**PD**	Incidents resulting in asset/equipment damage only.

7	**Near Miss Cases**	$\text{Absolute Case Count}$	**NM**	Potential hazards with no actual injury or loss.
8	**Hazard Observation Cases**	$\text{Absolute Case Count}$	**HZ**	Identified unsafe behaviors or physical conditions.

5.3 Technical Specifications for Data Collection

1. Total Labor Hours Worked (Denominator): Represents the consolidated sum of Direct Employee Hours and Contractor/Subcontractor Hours within the specified Date Range.
2. Standard Classifications:
 - Classifications 1 to 5 require dynamic rate calculations (normalized per 1M hours).
 - Classifications 6 to 8 represent simple integer counters (Case Numbers) and do not utilize the labor hours denominator.
3. No-Activity Safety Baseline: If Total Labor Hours Worked is 0, the rates default to `0.00` with a warning flag instead of throwing division-by-zero exceptions.

6. Non-Functional & Technical Requirements

6.1 Performance and Scalability

- 🔧 ½ SLA on Extraction: Large-scale raw data dumps encompassing up to 100,000 historical rows must process and start downloading within 5 seconds.
- 🔧 ½ On-the-Fly Generation: PDFs and Charts must be generated server-side or via optimized client-side libraries (such as `jspdf` or `xlsx` with memory-efficient streams) without freezing or locking the browser tab.

6.2 Data Security & Auditing

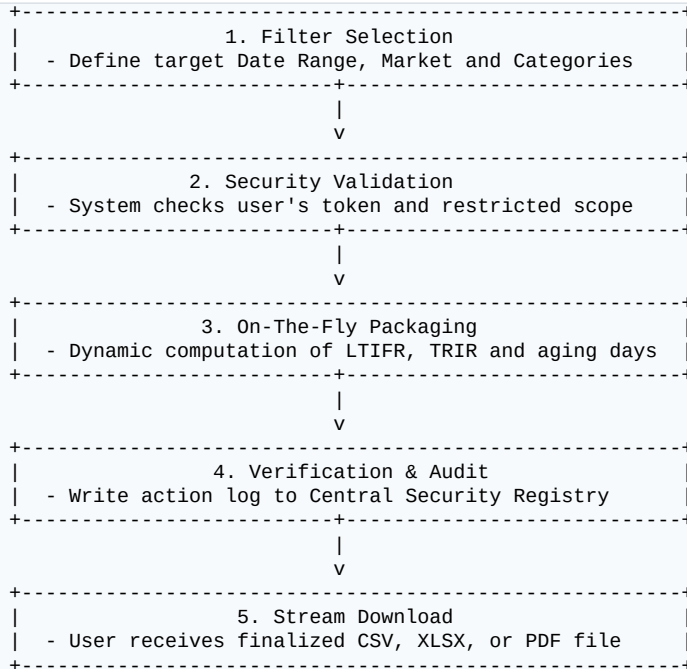
- 🔧 ½ Audit Logging: Every extraction activity must append a record to the central system audit log containing:
 - `User\_ID` & `Role`
 - `Action` (e.g. `EXPORT\_EXCEL\_CAPA\_ROSTER`, `EXPORT\_PDF\_EXECUTIVE\_SUMMARY`)
 - `Filters\_Applied` (e.g. `Country=Vietnam; Dates=2026-01-01\_to\_2026-06-30`)
 - `Rows\_Extracted\_Count`
- 🔧 ½ Data Encapsulation: Restrict front-end source maps from exposing any un-redacted datasets for unauthorized countries in the client-side state.

6.3 Compliance and Usability

- 🔧 ½ Accessibility: PDF documents should pass basic screen-reader compliance tests (contrast ratios of text to backgrounds must meet WCAG 2.1 AA standards).
- 🔧 ½ Mobile Friendliness: In mobile view, the extraction modal or settings drawer must be perfectly responsive, stacked vertically, with large click targets (minimum 44x44px for export buttons).

7. Workflow Acceptance Criteria

To declare the Report Extraction feature ready for production, it must successfully pass the following validation steps:



- 1. Test Case 1 (Superuser Scope):** Triggering a regional export correctly returns a aggregate summary table combining Thailand, Vietnam, and Malaysia with accurate mathematical weighting.
- 2. Test Case 2 (Reporter Isolation):** A user simulated with a "Reporter" role attempts to download Thailand data. The browser successfully receives the file containing *only* Thailand, with columns containing full worker names redacted/replaced with `[CONFIDENTIAL]`.
- 3. Test Case 3 (Zero Records Fallback):** If a selected date range has zero incident reports and zero working hours recorded, the export yields a clean, friendly notification warning instead of returning empty Excel errors.